# Dubbing QC

Dub vs original version — QC report · alpha

2026-11-01 14:47

## Dub

| | |
|---|---|
| **Languages** | English (OV) → dub |
| **Dub (checked)** | Dub — split-mono 5.1 |
| **Original version** | Original Version |
| **Programme length** | 00:55:19:03 (3319.12 s) |
| **Session start** | 07:00:00:00 @ 25 fps (the dub's BWF TimeReference) |
| **OV language** | English (en-US) — language checks NOT RUN (see below) |
| **Delivery profile** | Measured defaults |
| **Thresholds** | measured defaults |
| **Analysis time** | 20.1 s |

## Summary

76 findings. Timecodes are session timecode of the dub. Each finding is also exported as a DAW marker.

| | | | | | |
|---|---|---|---|---|---|
| 9 Missing line | 20 Missing effort | off Laugh missing | off Walla missing | 10 Extra in dub | off OV language |
| 5 Sync | 1 Cut-off line | 5 Click / pop | 22 Reverb differs | 1 Pan differs | 3 EQ differs |
| 0 LFE differs | | | | | |

Missing efforts are ranked, not judged: the 20 most audible are listed; 373 more moments where the dub is silent (mostly breaths) are exported as separate review markers (….effort-review.csv / .mid) to step through against picture — which of them a QC operator flags cannot be told from the audio. Extra sounds in the dub: the 10 most audible are listed, 4 quieter ones are not. Not checked: lip-sync against picture (the sync check compares against the OV audio only). Level jumps vs the OV (≥7 dB against the neighbouring lines): 37 found, exported as level review markers (….level-review.csv / .mid), not listed — most are intended performance differences, and the audio cannot tell which. OV audio reused in the dub (kept songs, walla, breaths; also when slipped by up to 100 ms; words heard are shown, but sung lyrics count as words too): 8 stretches, exported as OV audio review markers (….ov-audio-review.csv / .mid), not listed — usually intended; listen for an undubbed line.

## File QC

Format, length, loudness, peaks and channel layout of both versions. 0 fail, 0 to check. Loudness of a dialogue stem is reported, not judged: the delivery target applies to the printmaster.

| Check | OV | Dub | Result | Note |
|---|---|---|---|---|
| Sample rate | 48000 Hz | 48000 Hz | Pass | |
| Bit depth | 24-bit | 24-bit | Pass | |
| Channels | L R C LFE Ls Rs | L R C LFE Ls Rs | Pass | |
| Channel files same length | yes | yes | Pass | |
| Programme length | 00:55:19:03 | 00:55:19:03 | Pass | identical to the sample |
| File names: one name for all channels | yes | yes | Pass | |
| File name: frame rate | — | — | Info | no frame rate in either name |

| Check | OV | Dub | Result | Note |
|---|---|---|---|---|
| File name: convention | — | — | Info | the OV's name has no LANG-51 field convention; names not compared |
| Embedded start TC (BWF) | 00:00:00:00 | 07:00:00:00 | Info | different session placement; the checks align the versions from the first sample |
| Integrated loudness (BS.1770) | -26.8 LUFS | -25.3 LUFS | Info | dub +1.5 LU vs OV; a dialogue stem has no target of its own |
| Loudness range | 15.5 LU | 13.1 LU | Info | |
| True peak | -4.9 dBTP | -2.5 dBTP | Pass | EBU R128 ceiling –1 dBTP |
| Sample peak | -4.9 dBFS | -2.5 dBFS | Info | |
| Clipping (≥3 samples at full scale) | 0 | 0 | Pass | |
| LFE in a dialogue stem | empty | empty | Pass | |
| Dialogue in C (loudest channel) | C | C | Pass | |
| DC offset (worst channel) | -99.9 dBFS | -93.5 dBFS (C) | Pass | |
| L/R correlation | -0.00 | +0.07 | Pass | |
| Room tone between lines | room tone -60 dB | digital silence in 57% of 604 gaps, otherwise -82 dB | Info | Masked by the M&E in the mix; matters only where the delivery spec requires fill in the dialogue stem. |

## Findings

| # | Session TC | Length | Issue | Severity | Measurement |
|---|---|---|---|---|---|
| 1 | 07:00:14:12 | 21.5 s | Echo Reverb | Major | side channels vs C: OV -39 dB, dub -74 dB (Ls+Rs -50/-77, L+R -40/-77) |
| 2 | 07:00:19:20 | 1.8 s | Out of Sync Section | Major | dub line start +600 ms, end +740 ms vs OV |
| 3 | 07:00:47:00 | 3.5 s | Out of Sync Section | Major | dub line start -710 ms, end -1080 ms vs OV |
| 4 | 07:01:03:16 | 0.2 s | Extra Dialog (not in OV) | Major | rank 8 of 14: dub sneeze (classifier 0.54), -4 dB vs median dub dialogue; OV quiet within ±0.5 s |
| 5 | 07:04:38:15 | 0.0 s | Click Pop Tick | Major | jump -23.9 dBFS, spike -26.4 dBFS, 105× its surroundings |
| 6 | 07:08:36:08 | 0.2 s | Missing Dialog (effort) | Blocker | rank 17 of 351: OV speech (classifier 0.67), +19 dB vs median dialogue, centre-heavy; dub silent |
| 7 | 07:08:54:09 | 0.4 s | Missing Dialog (effort) | Blocker | rank 5 of 351: OV speech (classifier 0.91), +19 dB vs median dialogue, centre-heavy; dub silent |
| 8 | 07:09:05:13 | 0.0 s | Click Pop Tick | Major | jump -24.3 dBFS, spike -24.9 dBFS, 68× its surroundings |
| 9 | 07:09:17:01 | 0.3 s | Missing Dialog (effort) | Blocker | rank 13 of 351: OV speech (classifier 0.43), +19 dB vs median dialogue, centre-heavy; dub silent |
| 10 | 07:10:02:12 | 3.5 s | EQ / Voice Treatment | Major | OV has 20 dB less lows (80–300 Hz) — lows dub–OV +20 dB, highs dub–OV -8 dB |
| 11 | 07:10:30:04 | 0.0 s | Click Pop Tick | Major | jump -29.2 dBFS, spike -27.4 dBFS, 243× its surroundings |
| 12 | 07:10:56:15 | 0.4 s | Extra Dialog (not in OV) | Major | rank 6 of 14: dub sneeze (classifier 0.70), -5 dB vs median dub dialogue; OV quiet within ±0.5 s |
| 13 | 07:12:42:19 | 1.5 s | Missing Dialog | Blocker | DUB silent 1.5 s (38 frames) differs of OV — not run — speech recognition is switched off (it is optional; nothing was sent anywhere) |

| # | Session TC | Length | Issue | Severity | Measurement |
|---|---|---|---|---|---|
| 14 | 07:14:23:00 | 3.5 s | Echo Reverb | Major | dub has reverb the OV lacks — side channels vs C: OV -38 dB, dub -16 dB (Ls+Rs -48/-25, L+R -38/-17) |
| 15 | 07:14:38:05 | 5.5 s | Out of Sync Section | Major | dub line start +1500 ms, end +1260 ms vs OV |
| 16 | 07:15:37:17 | 2.0 s | Missing Dialog (effort) | Blocker | rank 6 of 351: OV speech (classifier 0.73), +12 dB vs median dialogue, centre-heavy; dub silent |
| 17 | 07:16:31:05 | 1.3 s | Extra Dialog (not in OV) | Major | rank 1 of 14: dub speech (classifier 0.95), +5 dB vs median dub dialogue; OV quiet within ±0.5 s |
| 18 | 07:16:37:20 | 1.4 s | Missing Dialog (effort) | Blocker | rank 11 of 351: OV speech (classifier 0.89), +12 dB vs median dialogue, centre-heavy; dub silent |
| 19 | 07:16:44:24 | 0.6 s | Extra Dialog (not in OV) | Major | rank 10 of 14: dub sneeze (classifier 0.33), -11 dB vs median dub dialogue; OV quiet within ±0.5 s |
| 20 | 07:18:03:12 | 4.0 s | Echo Reverb | Major | dub has reverb the OV lacks — side channels vs C: OV -74 dB, dub -21 dB (Ls+Rs -77/-25, L+R -77/-24) |
| 21 | 07:18:17:12 | 10.5 s | Echo Reverb | Major | dub has reverb the OV lacks — side channels vs C: OV -74 dB, dub -20 dB (Ls+Rs -77/-24, L+R -77/-23) |
| 22 | 07:18:28:00 | 0.1 s | Missing Dialog (effort) | Blocker | rank 20 of 351: OV speech (classifier 0.67), +23 dB vs median dialogue, centre-heavy; dub silent |
| 23 | 07:19:01:12 | 9.0 s | Echo Reverb | Major | side channels vs C: OV -0 dB, dub -74 dB (Ls+Rs -15/-77, L+R -0/-77) |
| 24 | 07:19:31:22 | 3.5 s | Missing Dialog | Blocker | DUB silent 3.5 s (88 frames) differs of OV — not run — speech recognition is switched off (it is optional; nothing was sent anywhere) |
| 25 | 07:19:35:12 | 96.0 s | Echo Reverb | Major | side channels vs C: OV -19 dB, dub -74 dB (Ls+Rs -30/-77, L+R -19/-77) |
| 26 | 07:20:01:09 | 0.2 s | Missing Dialog (effort) | Blocker | rank 8 of 351: OV speech (classifier 0.89), +20 dB vs median dialogue, centre-heavy; dub silent |
| 27 | 07:21:06:07 | 1.4 s | Missing Dialog (effort) | Blocker | rank 15 of 351: OV speech (classifier 0.96), +11 dB vs median dialogue, centre-heavy; dub silent |
| 28 | 07:21:27:00 | 5.5 s | Echo Reverb | Major | side channels vs C: OV -30 dB, dub -74 dB (Ls+Rs -39/-77, L+R -30/-77) |
| 29 | 07:21:35:12 | 3.0 s | EQ / Voice Treatment | Major | OV has 27 dB less lows (80–300 Hz), OV has 26 dB less highs (5–12 kHz) — lows dub–OV +27 dB, highs dub–OV +26 dB |
| 30 | 07:21:43:12 | 3.0 s | EQ / Voice Treatment | Major | OV has 22 dB less lows (80–300 Hz) — lows dub–OV +22 dB, highs dub–OV +12 dB |
| 31 | 07:21:53:00 | 13.5 s | Echo Reverb | Major | dub has reverb the OV lacks — side channels vs C: OV -45 dB, dub -23 dB (Ls+Rs -55/-34, L+R -45/-24) |
| 32 | 07:22:15:00 | 13.0 s | Echo Reverb | Major | side channels vs C: OV -41 dB, dub -74 dB (Ls+Rs -56/-77, L+R -41/-77) |
| 33 | 07:22:36:02 | 2.1 s | Missing Dialog (effort) | Blocker | rank 9 of 351: OV speech (classifier 0.88), +11 dB vs median dialogue, centre-heavy; dub silent |
| 34 | 07:22:41:00 | 31.0 s | Echo Reverb | Major | side channels vs C: OV -42 dB, dub -74 dB (Ls+Rs -52/-77, L+R -43/-77) |
| 35 | 07:23:25:08 | 2.0 s | Missing Dialog | Blocker | DUB silent 2.0 s (50 frames) differs of OV — not run — speech recognition is switched off (it is optional; nothing was sent anywhere) |
| 36 | 07:23:51:12 | 3.0 s | Echo Reverb | Major | side channels vs C: OV -26 dB, dub -63 dB (Ls+Rs -38/-75, L+R -26/-63) |
| 37 | 07:25:41:02 | 2.0 s | Missing Dialog | Blocker | DUB silent 2.0 s (50 frames) differs of OV — not run — speech recognition is switched off (it is optional; nothing was sent anywhere) |

| # | Session TC | Length | Issue | Severity | Measurement |
|---|---|---|---|---|---|
| 38 | 07:25:43:02 | 1.5 s | Missing Dialog | Blocker | DUB silent 1.5 s (38 frames) differs of OV — not run — speech recognition is switched off (it is optional; nothing was sent anywhere) |
| 39 | 07:25:50:14 | 3.5 s | Missing Dialog | Blocker | DUB silent 3.5 s (88 frames) differs of OV — not run — speech recognition is switched off (it is optional; nothing was sent anywhere) |
| 40 | 07:25:54:14 | 2.0 s | Missing Dialog | Blocker | DUB silent 2.0 s (50 frames) differs of OV — not run — speech recognition is switched off (it is optional; nothing was sent anywhere) |
| 41 | 07:25:58:20 | 1.6 s | Missing Dialog (effort) | Blocker | rank 1 of 351: OV sneeze (classifier 0.83), +15 dB vs median dialogue, centre-heavy; dub silent |
| 42 | 07:26:50:11 | 0.2 s | Missing Dialog (effort) | Blocker | rank 14 of 351: OV speech (classifier 0.67), +19 dB vs median dialogue, centre-heavy; dub silent |
| 43 | 07:27:04:17 | 0.1 s | Missing Dialog (effort) | Blocker | rank 10 of 351: OV sigh (classifier 0.62), +23 dB vs median dialogue, centre-heavy; dub silent |
| 44 | 07:27:21:00 | 43.5 s | Echo Reverb | Major | side channels vs C: OV -30 dB, dub -63 dB (Ls+Rs -43/-74, L+R -30/-63) |
| 45 | 07:27:51:02 | 0.2 s | Missing Dialog (effort) | Blocker | rank 19 of 351: OV speech (classifier 0.97), +21 dB vs median dialogue, centre-heavy; dub silent |
| 46 | 07:28:49:03 | 0.5 s | Extra Dialog (not in OV) | Major | rank 3 of 14: dub breathing (classifier 0.81), -5 dB vs median dub dialogue; OV quiet within ±0.5 s |
| 47 | 07:31:26:06 | 0.0 s | Click Pop Tick | Major | jump -25.5 dBFS, spike -29.4 dBFS, 155× its surroundings |
| 48 | 07:31:26:11 | 1.2 s | Cut-off Dialog | Major | line stops at full level: 25–60 ms before the end it is +1.1 dB vs its speaking level (natural ends are > 7 dB down) |
| 49 | 07:31:37:00 | 8.5 s | Echo Reverb | Major | side channels vs C: OV -22 dB, dub -74 dB (Ls+Rs -42/-77, L+R -22/-77) |
| 50 | 07:31:39:11 | 0.6 s | Missing Dialog (effort) | Blocker | rank 3 of 351: OV screaming (classifier 0.44), +18 dB vs median dialogue, centre-heavy; dub silent |
| 51 | 07:31:56:00 | 4.5 s | Echo Reverb | Major | side channels vs C: OV -32 dB, dub -74 dB (Ls+Rs -53/-77, L+R -32/-77) |
| 52 | 07:32:10:12 | 4.0 s | Echo Reverb | Major | side channels vs C: OV -17 dB, dub -74 dB (Ls+Rs -35/-77, L+R -17/-77) |
| 53 | 07:32:11:09 | 2.8 s | Missing Dialog (effort) | Blocker | rank 18 of 351: OV speech (classifier 0.76), +8 dB vs median dialogue, centre-heavy; dub silent |
| 54 | 07:32:16:13 | 1.5 s | Missing Dialog | Blocker | DUB silent 1.5 s (38 frames) differs of OV — not run — speech recognition is switched off (it is optional; nothing was sent anywhere) |
| 55 | 07:32:40:00 | 5.0 s | Echo Reverb | Major | dub has reverb the OV lacks — side channels vs C: OV -41 dB, dub -18 dB (Ls+Rs -46/-21, L+R -43/-22) |
| 56 | 07:33:38:21 | 0.5 s | Extra Dialog (not in OV) | Major | rank 4 of 14: dub sigh (classifier 0.96), -6 dB vs median dub dialogue; OV quiet within ±0.5 s |
| 57 | 07:33:44:00 | 25.0 s | Echo Reverb | Major | dub has reverb the OV lacks — side channels vs C: OV -64 dB, dub -36 dB (Ls+Rs -68/-48, L+R -67/-36) |
| 58 | 07:34:14:00 | 2.6 s | Out of Sync Section | Major | dub line start +560 ms, end +520 ms vs OV |
| 59 | 07:34:17:21 | 0.0 s | Click Pop Tick | Major | jump -25.3 dBFS, spike -29.7 dBFS, 89× its surroundings |
| 60 | 07:35:29:00 | 3.5 s | Volume Pan Shift | Major | R–L balance: OV -3 dB, dub +11 dB |
| 61 | 07:35:42:00 | 14.5 s | Echo Reverb | Major | side channels vs C: OV -23 dB, dub -71 dB (Ls+Rs -51/-77, L+R -23/-73) |
| 62 | 07:37:58:00 | 3.3 s | Missing Dialog (effort) | Blocker | rank 16 of 351: OV speech (classifier 0.62), +8 dB vs median dialogue, centre-heavy; dub silent |

| # | Session TC | Length | Issue | Severity | Measurement |
|---|---|---|---|---|---|
| 63 | 07:38:09:11 | 5.7 s | Missing Dialog (effort) | Blocker | rank 2 of 351: OV sigh (classifier 0.78), +9 dB vs median dialogue, centre-heavy; dub silent |
| 64 | 07:39:50:21 | 1.5 s | Missing Dialog | Blocker | DUB silent 1.5 s (38 frames) differs of OV — not run — speech recognition is switched off (it is optional; nothing was sent anywhere) |
| 65 | 07:40:22:19 | 0.2 s | Missing Dialog (effort) | Blocker | rank 12 of 351: OV speech (classifier 0.95), +21 dB vs median dialogue, centre-heavy; dub silent |
| 66 | 07:40:59:09 | 0.3 s | Missing Dialog (effort) | Blocker | rank 4 of 351: OV speech (classifier 0.81), +22 dB vs median dialogue, centre-heavy; dub silent |
| 67 | 07:41:10:02 | 0.5 s | Extra Dialog (not in OV) | Major | rank 2 of 14: dub sigh (classifier 0.99), +0 dB vs median dub dialogue; OV quiet within ±0.5 s |
| 68 | 07:41:11:00 | 5.0 s | Echo Reverb | Major | side channels vs C: OV -38 dB, dub -74 dB (Ls+Rs -48/-77, L+R -39/-77) |
| 69 | 07:44:15:04 | 0.7 s | Extra Dialog (not in OV) | Major | rank 7 of 14: dub sneeze (classifier 0.96), -8 dB vs median dub dialogue; OV quiet within ±0.5 s |
| 70 | 07:46:54:00 | 5.0 s | Echo Reverb | Major | dub has reverb the OV lacks — side channels vs C: OV -65 dB, dub -26 dB (Ls+Rs -68/-28, L+R -68/-30) |
| 71 | 07:49:36:00 | 56.0 s | Echo Reverb | Major | dub has reverb the OV lacks — side channels vs C: OV -74 dB, dub -29 dB (Ls+Rs -77/-33, L+R -77/-32) |
| 72 | 07:50:17:05 | 2.9 s | Out of Sync Section | Major | dub line start +930 ms, end +1420 ms vs OV |
| 73 | 07:51:22:02 | 2.2 s | Missing Dialog (effort) | Blocker | rank 7 of 351: OV speech (classifier 0.86), +11 dB vs median dialogue, centre-heavy; dub silent |
| 74 | 07:52:09:00 | 20.0 s | Echo Reverb | Major | dub has reverb the OV lacks — side channels vs C: OV -74 dB, dub -33 dB (Ls+Rs -77/-42, L+R -77/-34) |
| 75 | 07:52:57:02 | 0.6 s | Extra Dialog (not in OV) | Major | rank 5 of 14: dub breathing (classifier 0.86), -6 dB vs median dub dialogue; OV quiet within ±0.5 s |
| 76 | 07:52:58:00 | 0.9 s | Extra Dialog (not in OV) | Major | rank 9 of 14: dub breathing (classifier 0.55), -10 dB vs median dub dialogue; OV quiet within ±0.5 s |

## Rejected by the second pass

Missing-line candidates are checked with not run — speech recognition is switched off (it is optional; nothing was sent anywhere); rejected when pass 1 heard speech-like sound but no words were recognised. OV-language flags are rejected when the OV says different words there. 0 rejected, listed for transparency, not as markers. OV-language speech in the dub: not run — speech recognition is switched off (it is optional; nothing was sent anywhere).

## How each check works

Alpha: thresholds come from one episode, its client QC report and listening tests. Expect them to move as more episodes are checked. The numbers below are the measured defaults; this run used them unchanged.

| Check | Method |
|---|---|
| Missing Dialog | Pass 1: Apple's sound classifier hears at least ~1.5 s of speech-like sound in the OV inside a silence of the dub. Pass 2: on-device speech recognition on that OV stretch — ≥2 words confirms a line, 1 word is uncertain, none rejects it as background. |
| Missing Dialog (effort) | The OV has a centre-heavy vocal sound (≥15 dB over the side channels: a close-miked character, not background) where the dub is silent; the classifier must hear a voice. Moments are ranked by loudness × duration and only the top 20 are listed: quiet breaths usually rank lower and are counted, not listed. |
| Missing Dialog (laugh) | PANNs (Kong et al. 2020, AudioSet; weights CC BY 4.0) on each centre channel: five laughter classes per 0.32 s. The OV scores ≥0.1 and the dub ≤0.06 within ±0.5 s — also where the dub speaks, which the effort check cannot see. Listened on two episodes: 23 of the 25 strongest were real missing laughs, 2 unsure. A laugh the model does not know as laughter is missed. |

| Check | Method |
|---|---|
| Mix Balance (walla) | PANNs crowd classes (crowd, babble, chatter, cheering, applause) on the sum of L R C Ls Rs: the OV DX scores ≥0.07 and the dub ≤0.05, or under 40 % of the OV, within ±0.5 s for ≥0.6 s. The dub side is its dialogue PLUS any separate walla stem, rebuilt from the printmasters (dub PM − OV PM + OV DX); without printmasters the candidates go to a walla review file, since the dub's walla is often on another stem (63 DX-only stretches vs 16 with printmasters on the test episode). |
| Extra Dialog (not in OV) | The dub has a centre-heavy vocal sound (≥15 dB over its side channels; the classifier must hear a voice) where the OV sits at its room-tone floor with no activity within ±0.5 s — isolated, so the tails of longer dub lines do not count. Ranked by loudness × duration; the top 10 are listed, quieter ones (mostly breaths) are counted. |
| OV Audio in Dub | Pass 1: per second of dub dialogue, the correlation of the dub and OV centre-channel waveforms — two performances correlate near 0, the same recording near 1; ≥0.5 held ≥1 s, also searched ±100 ms for a slipped copy. Pass 2: on-device speech recognition in the OV language on the stretch shows any words heard (≥2 at mean confidence ≥0.6). Every stretch is a review marker, not a finding: kept songs, walla and breaths are usually intended, and sung lyrics are recognised as words too, so only listening decides. |
| Untranslated Dialog | Every dub line of ≥1 s (outside the OV-in-dub stretches) goes through Apple's on-device speech recogniser for the OV language; ≥2 words at mean confidence ≥0.6 flags OV-language speech in the dub. Pass 2: the OV is transcribed around the line (±1 s); a flag is kept only if the two share ≥2 words — an untranslated or re-recorded original line says what the OV says, a misheard dub line does not. Lines under 1 s are not judged. |
| Out of Sync Section | A dub line's start and end are both displaced from the OV line around it by ≥500 ms in the same direction. With printmasters: per 4 s window the dub printmaster is aligned to the OV printmaster (and the dub DX to the dub printmaster) on their shared M&E, to the millisecond; an overall offset is a File QC fail, and a stretch of ≥3 agreeing windows over ≥8 s at another offset is marked ("PM SYNC"). |
| Cut-off Dialog | A dub line whose sound is still at its own speaking level 25–60 ms before it stops (≤1 dB down): the edit chopped the tail. Natural line ends are more than 7 dB down by then (98 % of lines on the test episode). Provisional: two real cuts confirmed by ear so far. |
| Click Pop Tick | A sample-level spike: jump 30× the local signal, 5-sample median residual 50× its surroundings and ≥ −30 dBFS; any jump in digital silence; 0.2–50 ms dropouts to zero. |
| Clipping | Three or more consecutive samples at full scale (\|x\| ≥ 0.999) in any dub channel. Part of the file QC pass. |
| Echo Reverb | Where both versions have dialogue, the four side channels (L, R, Ls, Rs) relative to C. Missing: the OV has reverb and the dub is ≥20 dB lower and dry, or ≥23 dB lower. Added: the dub's sides are ≥20 dB above the OV's and at least −40 dB vs C (surround reverb where the OV is dry or mono). Held ≥3 s. |
| Volume Pan Shift | The R−L balance differs from the OV by ≥10 dB while the favoured side carries dialogue-level sound (≥ −20 dB vs C). Held ≥3 s. |
| EQ / Voice Treatment | Per 1 s of dialogue in both, the centre channel's lows (80–300 Hz vs 500–2000 Hz) and highs (5–12 kHz vs 2–4 kHz). A second ≥25 dB apart starts a candidate; the whole stretch (≥2 s) must still differ by ≥20 dB at its median: a phone/radio/band-limited treatment on one version only. Different voices normally differ by ±16 dB. |
| LFE | Per 0.5 s, the LFE channel of each dialogue stem; the dub's corrected for the stems' overall dialogue level (≤ ±6 dB). Marked where either LFE is active (≥ −70 dBFS) and the two differ by ≥10 dB — the dub's LFE missing, lower, extra or louder — held ≥0.5 s; also where ≥ −10 dB of the dub LFE's energy lies above 200 Hz (full-band sound routed to LFE) and the OV's does not. Tested on planted LFE only: the test episodes' DX stems carry no LFE at all. |
| Mix Balance | A dub line's level against the OV differs by ≥7 dB from its neighbouring lines. Exported as a separate level review marker file, not marked: on the test episode 2 of 13 heard were real, and nothing measured (size, length, K-weighted loudness, which version departs from its neighbours) separated them. |